

Mining Biosynthetic Gene Clusters in *Helicobacter pylori* Genomes Associated with Premalignant and Malignant Gastric Phenotypes: Preliminary Results and Next Steps

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Abstract. *Biosynthetic gene clusters (BGCs) encode microbial secondary metabolites, several of which have documented roles in tumor biology, yet BGC mining has never been applied to Helicobacter pylori in the context of gastric carcinogenesis. Building on an unpublished undergraduate pilot study by the same group, which mined BGCs from two metagenome-assembled H. pylori genomes, this work-in-progress applies the same funcscan-based strategy (anti-SMASH, DeepBGC, GECCO) to 554 preassembled H. pylori genomes from cancer and precancerous cohorts, followed by domain-based Gene Cluster Family (GCF) clustering and statistical testing (Fisher's exact test, CMH, and logistic regression, corrected for multiple testing across the full tested set). Preliminary results show a subtle but significant increase in BGC richness in cancer genomes, with no individual GCF reaching significance after correction, motivating the next steps described here.*

1. Introduction

Biosynthetic gene clusters (BGCs) are co-located sets of genes responsible for the production of microbial secondary metabolites. Genome-mining tools such as anti-SMASH and DeepBGC enable their identification and classification from genomic and metagenomic data [Blin et al. 2025, Hannigan et al. 2019]. Although BGCs have already been linked to antitumor agents in other bacterial genera, to our knowledge no study to date has directly investigated bacterial BGCs in the context of the gastric cancer microbiome; the existing literature on this topic focuses on broad-spectrum taxonomic and functional profiling, without incorporating the biosynthetic perspective [Coker et al. 2018, Noto and Peek 2017].

In an unpublished undergraduate thesis by this group¹, BGC mining was applied, as a pilot, to two metagenome-assembled *H. pylori* genomes (MAGs) recovered from gastric biopsies of two high-risk gastric cancer patients. Despite the very low fraction of recoverable microbial DNA in human samples (about 2.2% of filtered reads), 28 biosynthetic regions of interest were identified, including modules linked to the *de novo* lipogenesis pathway (acetyl-CoA carboxylase, beta-ketoacyl-ACP synthase, 3-oxoacyl-ACP

¹Lima, C. E. C. *Identificação de Biosynthetic Gene Clusters (BGCs) associados à carcinogênese em histopatologias gástricas*. Undergraduate thesis, Ribeirão Preto Medical School, University of São Paulo, 2025 (unpublished).

reductase) and genes involved in tRNA modification and capsular polysaccharide biosynthesis, several of which have documented parallels in the tumor progression literature. This pilot result motivated the present stage of the project, a Master’s thesis, which forgoes *de novo* metagenomic assembly – heavily limited at scale given the low bacterial coverage depth achievable in human tissue – in favor of mining BGCs from already-sequenced, publicly deposited *H. pylori* genomes, enabling a formal statistical comparison between clinical phenotypes with substantially greater sample size.

2. Methods

We obtained 554 preassembled *H. pylori* genomes from NCBI, 179 from gastric-cancer-related samples and 375 from precursor cases (chronic gastritis and intestinal metaplasia). BGC mining was performed with the `nf-core/funcscan` pipeline (antiSMASH 8.0.1, DeepBGC 0.1.31, and GECCO 0.9.10), with results aggregated by `com-BGC` into a single, harmonized table of Pfam domains per BGC [Ewels et al. 2020, Di Tommaso et al. 2017, Frangenberg et al. 2024].

Since BiG-SCAPE requires antiSMASH-formatted GenBank files – incompatible with DeepBGC and GECCO outputs – we implemented, using Python and Bash scripts, a tool-agnostic Gene Cluster Family (GCF) clustering workflow operating directly on the harmonized Pfam domain arrays. Prior to similarity computation, DeepBGC BGCs with probability < 0.50 were excluded, and BGCs without an assigned product class (unclassified DeepBGC predictions and GECCO “Unknown”) were retained as a separate class. Pairwise similarity followed an adaptation of [Robey et al. 2021] under two regimes, depending on whether a BGC’s backbone domain could be verified by `hmm-scan`. Backbone-verified classes comprised antiSMASH’s Terpene-precursor and Azole-containing-RiPP calls, together with – following empirical marker-domain discovery on this dataset – DeepBGC’s NRP, Saccharide, Polyketide, Terpene, and Polyketide-Terpene classes; collectively, these accounted for 43.1% of all Jaccard-qualifying BGC pairs. For BGCs in these backbone-verified classes,

$$\text{Score} = \sqrt{0.8 \times \text{sequence identity (hmmalign)} + 0.2 \times \text{Jaccard}};$$

for all remaining BGCs, $\text{Score} = \sqrt{0.2 \times \text{Jaccard}}$. Final GCF assignment used DBSCAN ($\epsilon = 0.56$, `min_samples = 2`, selected via an epsilon-sensitivity sweep) on the resulting distance matrix ($1 - \text{Score}$).

For statistical testing, every non-singleton GCF carried by at least 3 genomes (a criterion set *a priori*) was tested for association with clinical phenotype (cancer vs. non-cancer) using Fisher’s exact test, the Cochran–Mantel–Haenszel (CMH) test stratified by continent, and multivariable logistic regression (adjusted for continent, age, and total BGC richness per genome), with Benjamini–Hochberg (FDR) correction applied across the entire tested set [Benjamini and Hochberg 1995].

3. Preliminary Results

Funcscan mining across the 554 genomes yielded 2,078 BGC predictions before filtering (2,076 after), of which 1,828 (88.1%) were grouped into 135 non-singleton GCFs and 248 remained singletons. Scoring BGC similarity across three tools with no shared coordinate

system was a nontrivial challenge: without a reliable cross-tool identity anchor, predictions with unresolved backbone domains had to fall back on a Jaccard-only score, which under-penalized cross-tool, cross-class similarity and let a single dominant cluster absorb over a thousand unrelated BGCs. Extending hmmlalign-based sequence-identity scoring to additional product classes (see Methods) resolved most of this over-merging: the dominant GCF (GCF_0001) now comprises exactly 554 BGCs, corresponding one-to-one to the universal terpene-precursor locus detected by antiSMASH in virtually every genome. This GCF is carried by 179/179 cancer genomes and 373/375 non-cancer genomes; being essentially invariant with respect to phenotype, it cannot itself drive the group-level richness differences reported below, and it was, as expected, non-significant when tested for phenotype association (Fisher’s exact test, $p \approx 1$). The median size of the remaining GCFs was still only 3 BGCs (Figure 1), reflecting the atypical and restricted biosynthetic repertoire of *H. pylori* already documented in the pilot study.

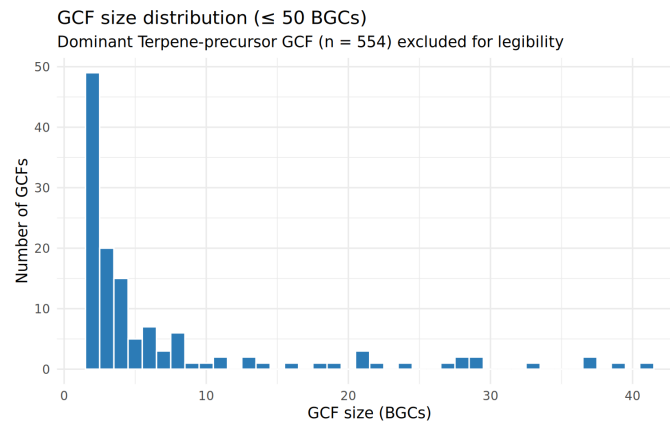


Figure 1. Distribution of non-singleton GCF sizes (≤ 50 BGCs shown for legibility; the dominant terpene-precursor GCF, 554 BGCs, is excluded).

BGC richness per genome was modestly, but significantly, higher in cancer genomes overall (median of 4 BGCs/genome vs. 3 in non-cancer; Wilcoxon $W = 40459.5$; $p = 5.96 \times 10^{-5}$; Figure 2) – a subtle difference in absolute terms, but a statistically robust one. Breaking this down by prediction tool: antiSMASH showed virtually identical distributions between Cancer and Non-cancer ($W = 33898.5$; $p_{adj} = 0.387$; ns), consistent with this tool typically detecting only 1–2 BGCs per genome; DeepBGC showed a significant difference ($W = 37563.0$; $p = 1.30 \times 10^{-5}$; $p_{adj} = 2.61 \times 10^{-5}$; ***); GECCO could not be evaluated due to its low prediction frequency.

GCF-level richness per genome also differed significantly by the Wilcoxon test ($W = 39892$; $p = 2.10 \times 10^{-4}$), even though the median was identical (3 GCFs/genome) in both groups. We flag this result with a caveat: because the Wilcoxon test is sensitive to distributional shape beyond the median (dispersion, ties, skew), a significant p -value alongside equal medians should not, on its own, be read as evidence of a clinically meaningful shift in typical GCF richness. We report it here as a distributional signal that warrants further characterization – for instance via an effect-size estimate or a closer look at the tails of the distribution – rather than as a straightforward “cancer genomes carry more GCFs” finding.

At the individual-GCF level, testing the 85 GCFs (out of 134 non-singleton GCFs

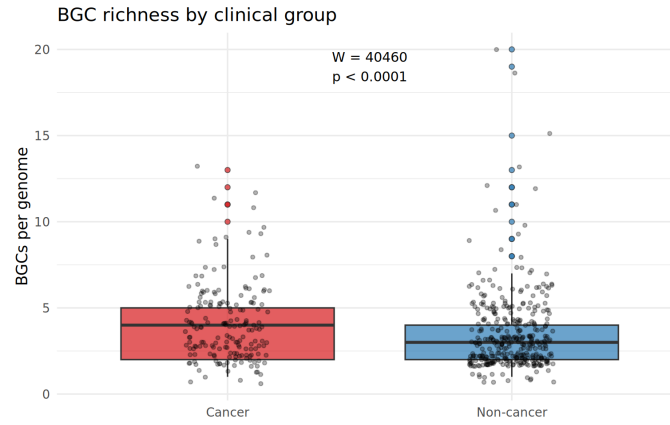


Figure 2. BGC richness per genome (Cancer vs. Non-cancer), Wilcoxon test.

in total) that met the pre-specified ≥ 3 -carrier-genome criterion, Fisher’s exact test with Benjamini–Hochberg correction across the full tested set found no GCF with $p_{adj} < 0.05$ or even $p_{adj} < 0.10$ (smallest observed p_{adj} : 0.189). Of these 85 eligible GCFs, 70 yielded a computable continent-stratified CMH statistic and all 85 yielded a computable multivariable logistic regression estimate; none reached adjusted significance by either method (0/70 and 0/85 respectively, $p_{adj} < 0.05$). In other words, despite the aggregate richness signal described above, no individual GCF carries a statistically robust association with cancer phenotype in this cohort once the number of GCFs actually tested is properly accounted for. Many of these 85 GCFs are supported by carrier counts close to the pre-specified minimum, which limits statistical power at the individual-family level; the absence of significant associations after correction should therefore be read as an absence of evidence for GCF-specific effects in this cohort, not as evidence that no such effects exist.

A quasi-Poisson model (BGC richness $\sim$ phenotype + age) showed additive, independent effects: IRR = 1.005 per year of age (95% CI: 1.001–1.008; $p = 0.0047$) and a borderline positive association with cancer phenotype (IRR = 1.119; 95% CI: 1.000–1.250; $p = 0.0487$). With its confidence interval’s lower bound touching unity, this cancer-phenotype effect is best read as suggestive rather than conclusive, and should be weighted accordingly relative to the more robust Wilcoxon signal reported above. No evidence of a phenotype $\times$ age interaction was found (likelihood-ratio test, $p = 0.864$). Spearman correlations between age and BGC richness were weak but significant overall ($\rho = 0.16$; $p = 1.14 \times 10^{-4}$) and in the non-cancer subgroup ($\rho = 0.15$; $p = 0.0048$), but not significant in the cancer subgroup alone ($\rho = 0.08$; $p = 0.305$); this does not contradict the absence of an interaction detected by the pooled model.

4. Discussion

Taken together, the preliminary results point to a subtle but statistically robust increase in overall BGC/GCF richness in cancer genomes (Wilcoxon tests on both metrics), reinforced – albeit more tentatively, given its confidence interval’s proximity to unity – by a borderline positive association from an independent quasi-Poisson model, while no individual GCF carries a significant association after appropriate correction for the number of GCFs tested. This is consistent with the hypothesis, already raised in the pilot study,

that *H. pylori* carries a restricted and highly conserved biosynthetic repertoire, in which phenotype-related effects may manifest more as quantitative differences in overall biosynthetic load than as the presence or absence of specific families. The breakdown by prediction tool (antiSMASH ns vs. DeepBGC significant) is reported here for methodological transparency, and it reinforces the value of combining rule-based and deep-learning-based mining approaches when working with an organism whose secondary metabolism is atypical [Zhu et al. 2025].

The resolution of the clustering challenge described in Preliminary Results also clarifies a methodological point: it was the extension of hmalign-based sequence-identity scoring to more product classes, rather than the epsilon-sensitivity sweep itself, that drove the improvement in GCF structure. This indicates that backbone-domain identity, when available, is the dominant signal for GCF granularity in this dataset, with the Jaccard term acting mainly as a fallback for tool-specific or unclassified predictions.

Several limitations should be considered when interpreting these results. First, the cohort is imbalanced (179 cancer vs. 375 non-cancer genomes), which, despite being accounted for in the stratified and adjusted models, reduces power to detect phenotype-specific effects at the individual-GCF level. Second, as with any study based on public genome assemblies, residual confounding cannot be fully ruled out: our models adjust for continent and age, but strain background, finer-grained geography, and clinical variables such as treatment history are not available in the associated metadata. Third, the tool-agnostic GCF clustering method introduced here is new and tailored to this cross-tool, cross-class prediction setting; it has not yet been benchmarked against an external gold standard such as BiG-SCAPE in a different organism, and its generalizability beyond *H. pylori* remains to be established.

5. Next Steps

Planned next steps for this Master’s project include: (i) correlating *H. pylori* phylogeny with the observed GCF repertoire; (ii) searching for a malignancy signature based on combinations of GCFs/BGCs, beyond the univariate GCF-by-GCF testing already performed; (iii) extending backbone-domain identity resolution to GECCO’s predictions (37 BGCs, currently Jaccard-only for lack of an extractable protein sequence source) and to DeepBGC’s remaining low-specificity classes (RiPP, Other); and (iv) discussing an experimental approach for functional validation of the most relevant BGCs in an animal model.

6. Conclusion

This work-in-progress extends, to a cohort of 554 public *H. pylori* genomes, the BGC mining approach validated as a pilot on two metagenome-assembled genomes. Preliminary results indicate a subtle but statistically robust signal of higher biosynthetic richness in genomes associated with gastric cancer – corroborated, more tentatively, by a borderline quasi-Poisson estimate of the same effect – and no individual GCF association surviving correction for multiple testing across the full eligible set. These findings, even though negative at the individual-GCF level, consolidate the clustering and pre-specified statistical testing methodology as a solid foundation for the phylogeny, combined malignancy signature, and functional validation analyses planned for the completion of the project.

7. Data and Code Availability

The 554 *H. pylori* genome accessions used in this study, together with associated metadata (phenotype, age, continent), are publicly available through NCBI. All analysis scripts, covering BGC mining, GCF clustering, and statistical testing, are available at https://github.com/carlosEcl06/mining_on_preassembled_genomes.

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